

Metric Spaces & Topology

Course Notes

Contents

1	Metric Spaces	2
1.1	Examples	2
2	Continuity in Metric Spaces	4
2.1	Open Balls	4
3	Open Sets	4
3.1	Continuity via Open Sets	5
4	Topological Spaces	5
4.1	Examples	6
4.2	Equivalent Metrics	6
4.3	Topologies on Finite Sets	7
4.4	Special Topologies	7
5	Basis for a Topology	7
6	Closed Sets, Closures, and Interiors	8
6.1	Limit Points	9
6.2	Boundary, Dense Sets, Interior	9
7	Convergence	10
7.1	Cauchy Sequences and Completeness	10
8	Continuous Functions Characterization	11

1 Metric Spaces

The idea of a metric and metric space are abstractions of the concept of distance in Euclidean space.

Definition 1.1 (Metric). A metric on a set X is a function $d : X \times X \rightarrow \mathbb{R}$ that satisfies the following conditions:

- (a) $d(x, y) \geq 0$ for all $x, y \in X$.
- (b) $d(x, y) = 0$ if and only if $x = y$.
- (c) $d(x, y) = d(y, x)$ for all $x, y \in X$ (Symmetry).
- (d) $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$ (Triangle Inequality).

If d is a metric on X , the ordered pair (X, d) is called a **Metric Space**.

Remark 1.1. Property (d) of a metric is an abstraction of the fact that the length of one side of a triangle is less than or equal to the sum of the lengths of the other two sides.

1.1 Examples

Example 1.1 (Usual Metric on $\mathbb{R}$). Let $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be defined by:

$$d(x, y) = |x - y|$$

This is called the *usual metric* on $\mathbb{R}$.

Example 1.2 (Usual Metric on $\mathbb{R}^2$). Let $d : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ where $x = (x_1, x_2)$ and $y = (y_1, y_2)$. Define:

$$d(x, y) = [(x_1 - y_1)^2 + (x_2 - y_2)^2]^{\frac{1}{2}}$$

Example 1.3 (Discrete Metric). Let X be a set. Define $d : X \times X \rightarrow \mathbb{R}$ by:

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

This is called the *discrete metric*.

Example 1.4 (Usual Metric on $\mathbb{R}^n$). Let $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$.

$$d(x, y) = \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{\frac{1}{2}}$$

This is called the *usual metric* (or Euclidean metric) for $\mathbb{R}^n$.

Remark 1.2. If we take $X = \mathbb{R}$ in Example 2 (Discrete), then we have two metrics on $\mathbb{R}$: the usual metric and the discrete metric.

Example 1.5 (Taxicab Metric). Let $n \in \mathbb{N}$ and define $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ by:

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

Then d is a metric on $\mathbb{R}^n$ and it is called the *Taxicab metric*.

Example 1.6 (Square Metric). Let $n \in \mathbb{N}$ and define a function $\rho : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ by:

$$\rho(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|, \dots, |x_n - y_n|\}$$

Then ρ is a metric on $\mathbb{R}^n$ and it is called the *Square metric*.

Verification of Square Metric Properties. Let $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$.

1. $\rho(x, y) \geq 0$ for all $x, y \in \mathbb{R}^n$.
2. $\rho(x, y) = \rho(y, x)$.
3. $\rho(x, y) = 0 \iff x = y$.
4. For each $i = 1, \dots, n$:

$$|x_i - z_i| \leq |x_i - y_i| + |y_i - z_i|$$

Therefore, by definition of ρ :

$$|x_i - z_i| \leq \rho(x, y) + \rho(y, z)$$

for each i . Hence, $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$.

□

Remark 1.3. For $\mathbb{R}^1$, the Euclidean metric, the Taxicab metric, and the Square metric are precisely the same.

Theorem 1.1 (Cauchy-Schwarz Inequality Context). Let $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n) \in \mathbb{R}^n$. Then:

$$|x \cdot y| \leq \|x\| \|y\|$$

Example 1.7 (l^2 Space). Let $X = l^2 = \{x = (x_1, \dots, x_n, \dots) : \sum_{i=1}^{\infty} |x_i|^2 < \infty\}$. Define:

$$d(x, y) = \left(\sum_{i=1}^{\infty} (x_i - y_i)^2 \right)^{\frac{1}{2}}$$

Then d is a well-defined metric on l^2 , called the l^2 -metric.

Example 1.8 (Integral Metric). Let $a, b \in \mathbb{R}$ with $a < b$, and let $X = C[a, b]$ denote the set of all continuous functions on $[a, b]$. Define a function $d : X \times X \rightarrow \mathbb{R}$ by:

$$d(f, g) = \int_a^b |f(x) - g(x)| dx$$

Then (X, d) is a metric space.

Proof. $d(f, g) \geq 0$ and $d(f, g) = d(g, f)$ are clear. Triangle inequality:

$$|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)|$$

Integrating both sides yields $d(f, h) \leq d(f, g) + d(g, h)$. It remains to show $d(f, g) = 0 \implies f \equiv g$. If $\int_a^b |f(x) - g(x)| dx = 0$ and $|f - g|$ is continuous and non-negative, then $|f(x) - g(x)| = 0$ for all x , so $f(x) = g(x)$. □

2 Continuity in Metric Spaces

Definition 2.1 (Continuous Function). Let (X, d) and (Y, ρ) be metric spaces. A function $f : X \rightarrow Y$ is **continuous at a point** $a \in X$ provided for each $\epsilon > 0$, there is a positive number $\delta > 0$ such that if $x \in X$ and $d(a, x) < \delta$, then $\rho(f(a), f(x)) < \epsilon$.

A function $f : X \rightarrow Y$ is **continuous** provided it is continuous at each point of X .

Example 2.1. Let (X, d) and (Y, ρ) be metric spaces, and let $y_0 \in Y$.

1. The identity map $i : X \rightarrow X$ defined by $i(x) = x$ is continuous.
2. The constant map $c : X \rightarrow Y$ defined by $c(x) = y_0$ for all $x \in X$ is continuous.

Proof for identity: Let $a \in X$. For any $\epsilon > 0$, choose $\delta = \epsilon$. If $d(a, x) < \delta$, then $d(i(a), i(x)) = d(a, x) < \epsilon$.

2.1 Open Balls

Definition 2.2. Let (X, d) be a metric space, let $\epsilon > 0$ be a positive real number, and let $x \in X$. The set:

$$B_d(x, \epsilon) = \{y \in X : d(x, y) < \epsilon\}$$

is called the ϵ -**ball** with center at x .

The definition of continuity in metric spaces can be rephrased in terms of open balls.

Theorem 2.1. Let (X, d) and (Y, ρ) be metric spaces. A function $f : X \rightarrow Y$ is continuous at a point $a \in X$ if and only if for each open ball $B_\rho(f(a), \epsilon)$, there is an open ball $B_d(a, \delta)$ such that:

$$f(B_d(a, \delta)) \subseteq B_\rho(f(a), \epsilon)$$

3 Open Sets

Definition 3.1. A subset U of a metric space (X, d) is **open** if for each $x \in U$, there is an open ball $B_d(x, \epsilon)$ such that $B_d(x, \epsilon) \subseteq U$.

Theorem 3.1 (Properties of Open Sets). *The open subsets of a metric space (X, d) have the following properties:*

- (a) X and $\emptyset$ are open sets.
- (b) The union of any collection of open sets is open.
- (c) The intersection of any finite collection of open sets is open.

Proof. (a) Let $x \in X$, and let $\epsilon > 0$. Then $B_d(x, \epsilon) \subseteq X$, hence X is open. The set $\emptyset$ is open because there is no $x \in \emptyset$ to violate the condition.

(b) Let $\{U_\alpha : \alpha \in \Lambda\}$ be a collection of open sets and let $U = \bigcup_{\alpha \in \Lambda} U_\alpha$. Let $x \in U$. Then there exists $\beta \in \Lambda$ such that $x \in U_\beta$. Since U_β is open, there exists an open ball $B_d(x, \epsilon)$ such that $B_d(x, \epsilon) \subseteq U_\beta$. Since $U_\beta \subseteq U$, we have $B_d(x, \epsilon) \subseteq U$. Thus U is open. $\square$

Remark 3.1. Notice that an arbitrary intersection of open sets need not be open. Let $U_n = (-\frac{1}{n}, \frac{1}{n})$ for $n \in \mathbb{N}$. Each U_n is an open subset of $\mathbb{R}$, but:

$$\bigcap_{n \in \mathbb{N}} U_n = \{0\}$$

And $\{0\}$ is not open in $\mathbb{R}$.

Lemma 3.1. *An open non-empty subset of $\mathbb{R}$ (with usual metric) is any set that can be expressed as the union of open intervals.*

Proof. If U is an open subset of $\mathbb{R}$, and let $x \in U$. Then there exists $\epsilon > 0$ such that $B_d(x, \epsilon) \subseteq U$. Note that $B_d(x, \epsilon) = (x - \epsilon, x + \epsilon)$. Therefore U can be expressed as a union of open intervals. $\square$

Example 3.1. Let X be a non-empty set, and let d be the discrete metric on X . Then for each $x \in X$ and each ϵ with $0 < \epsilon \leq 1$, $B_d(x, \epsilon) = \{x\}$. This implies $\{x\}$ is an open subset of (X, d) .

Remark 3.2. $\{0\}$ is not open in $\mathbb{R}$ with the usual metric, but it is open in $\mathbb{R}$ with respect to the discrete metric.

3.1 Continuity via Open Sets

Definition 3.2. Let $f : (X, d) \rightarrow (Y, \rho)$ be a function, and let $a \in X$. Then f is continuous at a if and only if for each open subset V of Y containing $f(a)$, there is an open subset U of X containing a such that $f(U) \subseteq V$.

4 Topological Spaces

The open sets in a metric space have properties listed in the previous theorem, and these properties are used to define a topology.

Definition 4.1 (Topology). A **topology** on a set X is a collection $\mathcal{T}$ of subsets of X having the following properties:

- (a) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.
- (b) If $U \in \mathcal{T}$ and $V \in \mathcal{T}$, then $U \cap V \in \mathcal{T}$.
- (c) If $U_\alpha \in \mathcal{T}$ for each α in an index set Λ , then $\bigcup_{\alpha \in \Lambda} U_\alpha \in \mathcal{T}$.

The members of $\mathcal{T}$ are called **open sets**.

Remark 4.1. Let $\mathcal{T}$ be a topology on a set X and let Λ be a finite set. If $U_\alpha \in \mathcal{T}$ for each $\alpha \in \Lambda$, then $\bigcap_{\alpha \in \Lambda} U_\alpha \in \mathcal{T}$.

Definition 4.2 (Topological Space). A topological space is a pair $(X, \mathcal{T})$, where X is a set and $\mathcal{T}$ is a topology on X .

4.1 Examples

Example 4.1 (Metric Topology). Let (X, d) be a metric space. The open subsets of (X, d) form a topology on X . This topology is called the topology on X **induced by d** . Thus every metric space is a topological space.

Definition 4.3 (Metriizable). A topological space $(X, \mathcal{T})$ is **metriizable** if there is a metric d on X such that the topology induced by d is $\mathcal{T}$.

Example 4.2 (Trivial Topology). Let X be a set with at least two members, and let $\mathcal{T} = \{\emptyset, X\}$ be the trivial topology on X . Then $(X, \mathcal{T})$ is **not metriizable**.

Proof. Let d be a metric on X and let $\mathcal{T}_d$ be the topology generated by d . We claim that $\mathcal{T} \neq \mathcal{T}_d$. Let a, b denote distinct members of X . Let $d(a, b) = r$. Then r is a positive number. Therefore $a \in B_d(a, r)$, but $b \notin B_d(a, r)$. This implies $B_d(a, r) \in \mathcal{T}_d$, but $B_d(a, r) \neq X$ and $B_d(a, r) \neq \emptyset$. Thus $\mathcal{T} \neq \mathcal{T}_d$. $\square$

Example 4.3 (Discrete Topology). Let X be a non-empty set, and let $\mathcal{T}$ denote the collection of all subsets of X ($\mathcal{T} = \mathcal{P}(X)$). Then $\mathcal{T}$ is a topology on X , called the **discrete topology**.

Remark 4.2. The discrete metric on a non-empty set induces the discrete topology on X .

Example 4.4. Let $\mathcal{T} = \{U \in \mathcal{P}(\mathbb{R}) : \text{for each } x \in U \text{ there is a positive number } \delta_x \text{ s.t. } (x - \delta_x, x + \delta_x) \subseteq U\}$. Then $\mathcal{T}$ is the topology on $\mathbb{R}$ induced by the usual metric.

Example 4.5. Let $\mathcal{T} = \{U \in \mathcal{P}(\mathbb{R}^2) : \text{for each } (a, b) \in U \text{ there is a positive number } \delta \text{ s.t. } \{(x - a)^2 + (y - b)^2 < \delta^2\} \subseteq U\}$. This is the topology on $\mathbb{R}^2$ induced by the usual metric.

4.2 Equivalent Metrics

Two metrics d and ρ on a set X induce the same topology on X provided that a subset of X is open in (X, d) if and only if it is open in (X, ρ) .

Example 4.6. The topology generated by the square metric ρ on $\mathbb{R}^2$ is the usual topology (generated by usual metric d).

$$\rho(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}$$

$$B_\rho(x, \epsilon) = \{y \in \mathbb{R}^2 : \rho(x, y) < \epsilon\}$$

Proof. Let d be the usual metric on $\mathbb{R}^2$. Let U be an open subset of $(\mathbb{R}^2, \rho)$, and let $x \in U$. Then there exists a positive number ϵ s.t. $B_\rho(x, \epsilon) \subseteq U$. Since $\rho(a, b) \leq d(a, b)$, we have $B_d(x, \epsilon) \subseteq B_\rho(x, \epsilon)$. Hence U is open in $(\mathbb{R}^2, d)$.

Conversely, let U be open in $(\mathbb{R}^2, d)$ and let $x \in U$. There exists $\epsilon > 0$ s.t. $B_d(x, \epsilon) \subseteq U$. Since $d(a, b) \leq \sqrt{2}\rho(a, b)$ for all $a, b \in \mathbb{R}^2$:

$$B_\rho(x, \frac{\epsilon}{\sqrt{2}}) \subseteq B_d(x, \epsilon) \subseteq U$$

Thus U is open in $(\mathbb{R}^2, \rho)$. $\square$

4.3 Topologies on Finite Sets

Let $X = \{1, 2, 3\}$. Some topologies on X :

- $\mathcal{T}_1 = \{\{1\}, \emptyset, X\}$
- $\mathcal{T}_2 = \{\{1, 2\}, \emptyset, X\}$
- $\mathcal{T}_3 = \{\{1\}, \{2\}, \{1, 2\}, \emptyset, X\}$
- $\mathcal{T}_4 = \{\{1\}, \{2, 3\}, \emptyset, X\}$
- $\mathcal{T}_5 = \{\{1, 2\}, \{2, 3\}, \{2\}, \emptyset, X\}$
- $\mathcal{T}_6 = \{\{1\}, \{1, 2\}, \emptyset, X\}$
- $\mathcal{T}_7 = \{\{1, 2\}, \{2, 3\}, \emptyset, X\}$ (Not a topology since intersection $\{2\}$ is missing).

Remark 4.3. There are many topologies on a set. If $\mathcal{T}_1$ and $\mathcal{T}_2$ are topologies on a set X and $\mathcal{T}_1 \subseteq \mathcal{T}_2$, then $\mathcal{T}_2$ is **finer** (stronger, larger) than $\mathcal{T}_1$, and $\mathcal{T}_1$ is **weaker** (coarser) than $\mathcal{T}_2$.

4.4 Special Topologies

Definition 4.4 (Finite Complement Topology). Let X be a set and let $\mathcal{T} = \{U \in \mathcal{P}(X) : U = \emptyset \text{ or } X \setminus U \text{ is finite}\}$. This is called the **finite complement topology**.

Remark 4.4. If X is finite, then the finite complement topology on X is the discrete topology.

Finite Complement Topology on $\mathbb{R}$: $\mathcal{T} = \{U \subseteq \mathbb{R} : \mathbb{R} \setminus U \text{ is finite, or } U = \emptyset\}$. Open sets include:

- $\mathbb{R} \setminus \{1\}$
- $\mathbb{R} \setminus \{1, 2, 3\}$
- All sets missing only finitely many points.

For $a \in \mathbb{R}$, the smallest open neighborhood of a is $\mathbb{R} \setminus F$ for some finite set $F \subseteq \mathbb{R} \setminus \{a\}$.

Definition 4.5 (Countable Complement Topology). Let X be a set and let $\mathcal{T} = \{U \in \mathcal{P}(X) : U = \emptyset \text{ or } X \setminus U \text{ is countable}\}$. This is called the **countable complement topology**.

5 Basis for a Topology

A topology set can be a complicated collection of subsets of a set X , and it can be difficult to describe the entire collection. In most cases one describes a subcollection that "generates" the topology. One such subcollection is called a Basis.

Definition 5.1 (Basis). Let $(X, \mathcal{T})$ be a topological space. A basis for $\mathcal{T}$ is a subcollection $\mathcal{B}$ of $\mathcal{T}$ with the property that if $U \in \mathcal{T}$, then $U = \emptyset$ or there is a subcollection $\mathcal{B}'$ of $\mathcal{B}$ such that $U = \bigcup \{B : B \in \mathcal{B}'\}$.

Example 5.1. 1. $\mathcal{B}$ = collection of all open intervals is a basis for the usual topology on $\mathbb{R}$.

2. If X is a set, then $\mathcal{B} = \{\{x\} : x \in X\}$ is a basis for the discrete topology on X .

3. Let (X, d) be a metric space. Then $\mathcal{B} = \{B(x, \epsilon) : x \in X, \epsilon > 0\}$ is a basis for the topology generated by d .

Example 5.2 (Counter-example). Let $X = \{1, 2, 3\}$ and $\mathcal{B} = \{\{1, 2\}, \{2, 3\}, X\}$. Suppose $\mathcal{B}$ is a basis for a topology $\mathcal{T}$. Then by definition $\mathcal{B} \subseteq \mathcal{T}$. Hence $\{1, 2\}, \{2, 3\} \in \mathcal{T} \implies \{1, 2\} \cap \{2, 3\} = \{2\} \in \mathcal{T}$. But $\{2\} \notin \mathcal{B}$ and no subcollection $\mathcal{B}'$ of $\mathcal{B}$ satisfies $\{2\} = \bigcup\{B : B \in \mathcal{B}'\}$. Hence $\mathcal{B}$ is not a basis for $\mathcal{T}$.

6 Closed Sets, Closures, and Interiors

Definition 6.1 (Closed Set). A subset A of a topological space $(X, \mathcal{T})$ is **closed** provided its complement, $X \setminus A$, is open.

Example 6.1. 1. A closed interval $[a, b]$, $a < b$ in $\mathbb{R}$ is closed with respect to the usual topology on $\mathbb{R}$, since $\mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, \infty)$ is open.

2. $\{a\}$ is closed with respect to the usual topology on $\mathbb{R}$, because $\mathbb{R} \setminus \{a\} = (-\infty, a) \cup (a, \infty)$.

3. With respect to the finite complement topology on X , a proper subset of X is closed if and only if it is a finite set.

Example 6.2 (Metric Space Closed Ball). If (X, d) is a metric space, $x \in X$, and $\epsilon > 0$, then $A = \{y \in X : d(x, y) \leq \epsilon\}$ is a closed subset of X .

Proof. To show A is closed, we show that $X \setminus A$ is open. Let $y \in X \setminus A$ and let $r_y = d(x, y)$. Then $r_y > \epsilon$. Let $\delta_y = r_y - \epsilon$. This is a positive number. Thus $B(y, \delta_y)$ is an open set. Claim: $B(y, \delta_y) \subseteq X \setminus A$. Suppose $z \in B(y, \delta_y)$. Then $d(y, z) < \delta_y$. Now $d(x, z) + d(z, y) \geq d(x, y)$.

$$d(x, z) \geq d(x, y) - d(z, y) > d(x, y) - \delta_y = r_y - \delta_y = \epsilon$$

$d(x, z) > \epsilon$ implies $z \notin A$. Therefore $B(y, \delta_y) \subseteq X \setminus A$. □

Theorem 6.1. Let $(X, \mathcal{T})$ be a topological space. Then the following conditions hold:

(a) X and $\emptyset$ are closed sets.

(b) If $\mathcal{A}$ is a finite family of closed subsets of X , then $\bigcup\{A : A \in \mathcal{A}\}$ is a closed set.

(c) If $\mathcal{A}$ is a family of closed subsets of X , then $\bigcap\{A : A \in \mathcal{A}\}$ is a closed set.

Definition 6.2 (Closure). The closure $\bar{A}$ of a subset A of a topological space $(X, \mathcal{T})$ is the intersection of the closed sets that contain A .

$\bar{A}$ is the "smallest" closed set that contains A . i.e., if B is any closed set s.t. $A \subseteq B$, then $\bar{A} \subseteq B$.

Theorem 6.2. Let A be a subset of a topological space $(X, \mathcal{T})$, and let $x \in X$. Then $x \in \bar{A}$ if and only if every neighborhood of x has a non-empty intersection with A .

Proof. Suppose $x \notin \bar{A}$. Then there is a closed set C such that $A \subseteq C$ and $x \notin C$. $X \setminus C$ is a neighborhood of x such that $(X \setminus C) \cap A = \emptyset$. Conversely, suppose there is a neighborhood U of x such that $U \cap A = \emptyset$. Then $X \setminus U$ is a closed set s.t. $A \subseteq X \setminus U$ and $x \notin X \setminus U$. Hence $x \notin \bar{A}$. $\square$

Remark 6.1. If (X, d) is a metric space and $A \subseteq X$, then it follows from the above theorem that for each positive ϵ , $B(x, \epsilon) \cap A \neq \emptyset$.

6.1 Limit Points

Definition 6.3. Let A be a subset of a topological space $(X, \mathcal{T})$. A point $x \in X$ is a **limit point** of A provided every neighborhood of x contains a point of A that is different from x . i.e., $U \cap (A \setminus \{x\}) \neq \emptyset$. Let A' denote the set of all limit points of A .

Example 6.3. With respect to the usual topology on $\mathbb{R}$, if $a, b \in \mathbb{R}$ and $a < b$, then every $x \in [a, b]$ is a limit point of each of $[a, b]$, $[a, b)$, $(a, b]$.

Theorem 6.3. Let A be a subset of a topological space $(X, \mathcal{T})$. Then $\bar{A} = A \cup A'$.

Proof. Let $x \in A \cup A'$. If $x \in A$, then if U is a nbhd of x , $U \cap A \neq \emptyset$. If $x \in A'$, then every nbhd of x contains a point of A different from x . In either case $x \in \bar{A}$. So $A \cup A' \subseteq \bar{A}$. Suppose $x \in \bar{A}$. If $x \notin A$, then every nbhd of x contains a point of A that is different from x (since it contains a point of A and $x \notin A$). $\implies x \in A'$. Thus $\bar{A} \subseteq A \cup A'$. $\square$

Theorem 6.4. Let A be a subset of a topological space $(X, \mathcal{T})$. Then A is closed if and only if $A' \subseteq A$.

6.2 Boundary, Dense Sets, Interior

Definition 6.4 (Boundary). Let A be a subset of a topological space $(X, \mathcal{T})$. A point $x \in X$ is a **boundary point** of A provided $x \in \bar{A} \cap X \setminus A$. The boundary of A , $bd(A)$, is the set of all boundary points of A .

Definition 6.5 (Dense). A subset A of a topological space $(X, \mathcal{T})$ is **dense** in X provided $\bar{A} = X$.

Definition 6.6 (Separable). If X has a countable dense subset, then $(X, \mathcal{T})$ is a **separable space**.

Example 6.4. 1. Let $(\mathbb{R}, \mathcal{T}_0)$ be the usual topology on $\mathbb{R}$. Then the set of rational numbers is a countable dense subset, hence $(\mathbb{R}, \mathcal{T}_0)$ is a separable space.

2. Let $\mathcal{T}$ be the finite complement topology on $\mathbb{R}$. Then every infinite subset is dense in $\mathbb{R}$.

Proof for (2): Let A be an infinite subset of $\mathbb{R}$. Let $x \in \mathbb{R}$ and let U be a nbhd of x . Since $\mathbb{R} \setminus U$ is finite, U must contain all but a finite number of members of A . In particular, $A \cap U \neq \emptyset$. Therefore $x \in \bar{A}$ and hence $\bar{A} = \mathbb{R}$.

Definition 6.7 (Interior). The interior, $int(A)$, of a subset A of a topological space $(X, \mathcal{T})$ is the union of the open sets that are subsets of A .

Definition 6.8 (Interior Point). Let A be a subset of a topo. space $(X, \mathcal{T})$ and let $x \in A$. Then x is an interior point if there is a nbhd U of x s.t. $U \subseteq A$.

7 Convergence

Recall: $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at a point x in $\mathbb{R}$ iff whenever $\{x_n\}$ is a sequence in $\mathbb{R}$ that converges to x , then $\{f(x_n)\}_n$ converges to $f(x)$.

Definition 7.1 (Convergence in Topological Space). A sequence $\{x_n\}$ of points in a topological space $(X, \mathcal{T})$ converges to a point x in X provided that for each neighborhood U of x there is a natural number N s.t. $x_i \in U$ for all $i \geq N$.

Example 7.1. 1. Let $\mathcal{T}_0$ be the trivial topology on a set X . Let $\{x_n\}$ be a sequence in X and let x be any member of X . Then $x_n \rightarrow x$. Indeed, let U be a nbhd of x . Since $\mathcal{T}_0$ is trivial topology and $X \in \mathcal{T}$, $U = X$. Therefore $x_n \in U$ for every $n \in \mathbb{N}$.

2. Let $\mathcal{T}$ be the countable complement topology on $\mathbb{R}$, and let A be the set of irrational numbers. Then $0 \in \bar{A}$, but there is no seq in A that converges to 0.

Proof for (2): Let U be a nbhd of 0. Then $\mathbb{R} \setminus U$ is countable. Thus, since A is uncountable, $A \not\subseteq \mathbb{R} \setminus U$. Therefore $U \cap A \neq \emptyset$, so $0 \in \bar{A}$. Let $\{x_n\}$ be any seq in A . Then $\mathbb{R} \setminus \{x_n\}_{n \in \mathbb{N}}$ is a nbhd of 0 that does not contain any point of $\{x_n\}$. Therefore $\{x_n\}_n$ does not converge to 0.

7.1 Cauchy Sequences and Completeness

Definition 7.2 (Cauchy Sequence). A seq. $\{x_n\}_n$ in a metric space (X, d) is a **Cauchy seq.** provided for each $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. if $m, n \geq N$ then $d(x_m, x_n) < \epsilon$.

Theorem 7.1. Let (X, d) be a metric space and let $\{x_n\}_n$ be convergent in X . Then $\{x_n\}_n$ is a Cauchy sequence.

Proof. Let $x_n \rightarrow x$. Let $\epsilon > 0$. Since $x_n \rightarrow x$, $\exists N$ s.t. for $n \geq N$, $d(x_n, x) < \frac{\epsilon}{2}$. Suppose $m, n \geq N$. Then:

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

□

Example 7.2. Let d be the usual metric on $\mathbb{R}$, let $X = \mathbb{R} \setminus \{0\}$. Is every Cauchy seq. convergent in (X, d) ? Let $x_n = \frac{1}{2^n}$. Then $\{x_n\}$ is Cauchy in (X, d) , yet it does not converge. (The limit $0 \notin X$).

Definition 7.3 (Complete). A metric space (X, d) is **complete** provided that every Cauchy sequence in X converges.

Theorem 7.2. Let (X, d) be a metric space such that every Cauchy seq in X has a convergent subsequence. Then (X, d) is complete.

Proof. Let $\{x_n\}$ be a Cauchy sequence, and let $\{x_{n_i}\}$ be a subsequence of $\{x_n\}$ that converges to x . Claim: $x_n \rightarrow x$. Let $\epsilon > 0$. $\exists N$ s.t. if $m, n \geq N$, then $d(x_m, x_n) < \frac{\epsilon}{2}$. Since $x_{n_i} \rightarrow x$, there is a natural number i_p s.t. $i_p > N$ and $d(x_{n_{i_p}}, x) < \frac{\epsilon}{2}$. Then for $n \geq N$:

$$d(x_n, x) \leq d(x_n, x_{n_{i_p}}) + d(x_{n_{i_p}}, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

□

Remark 7.1. If d is the usual metric on $\mathbb{R}$, then $(\mathbb{R}, d)$ is complete.

8 Continuous Functions Characterization

Recall:

Theorem 8.1. *Let $(X, \mathcal{T})$ and $(Y, \mathcal{U})$ be topological spaces and let $f : X \rightarrow Y$. Then the following statements are equivalent:*

- (a) *f is continuous.*
- (b) *If $V \in \mathcal{U}$, then $f^{-1}(V) \in \mathcal{T}$ (Inverse image of open is open).*
- (c) *If C is a closed subset of $(Y, \mathcal{U})$, then $f^{-1}(C)$ is a closed subset of $(X, \mathcal{T})$.*

Proof. (b) $\Rightarrow$ (c): Let C be a closed subset of $(Y, \mathcal{U})$. Then $Y \setminus C \in \mathcal{U}$. By (b), $f^{-1}(Y \setminus C) \in \mathcal{T}$. Now, $f^{-1}(C) = X \setminus f^{-1}(Y \setminus C)$. Therefore $f^{-1}(C)$ is a closed subset of $(X, \mathcal{T})$. $\square$